

AIDEN AYSUN

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Education

Massachusetts Institute of Technology

Bachelor of Science in Artificial Intelligence and Decision Making

Aug. 2026 – Present

Cambridge, MA

Oviedo High School

Co-Valedictorian of class of 566

Aug. 2022 – May 2026

Oviedo, FL

Relevant Coursework

- Physics Mechanics
- Physics E&M
- Multivariable Calculus
- Differential Equations

Experience

Machine Learning Undergraduate Researcher

Massachusetts Institute of Technology

Sep 2026 – Present

Cambridge, MA

- Currently developing a machine learning object detection model for the management of fisheries and aquaculture hatcheries.
- Specific responsibilities lie in developing a wrapper for a GUI, developing physics informed neural networks, and creating accurate training datasets.

Projects

Direct Tutor | *Node Modules, React, PostgreSQL, AWS, Stripe*

Jun. 2026 – Present

- Currently developing a full-stack application to match students with tutors based on specific user information.
- The frontend is built using React and Vite, and the backend is built using Node.js. App data is managed using PostgreSQL queries in Supabase. Payments are completed through Stripe with webhooks. Automated emailing is completed by using AWS SES via the AWS CLI on Linux.

Spectral Sensing Research | *Cosine Similarity, Temperature, Softmax Function, NumPy*

Jun. 2025 – Oct. 2025

- Co-authored a research paper with the goal of designing a robust color detection system under varying ambiance.
- Transformed the cosine similarity data, using a softmax function, into probability distributions in order to accurately compare the cosine similarity data with the probability distributions produced by the random forest model.
- Designed the Arduino firmware in order to log and plot the readings from 10 channels of the spectral sensor.
- Integrated a secure setup between the hardware, the Adafruit AS7341 spectral sensor, and Arduino Nano to ensure accurate spectral readings.

YouTube Rewind Controller | *JavaScript, CSS, HTML*

July 2025

- Published a Chrome Extension, built using Manifest V3, that adds a hotkey for continuous rewind playback of YouTube videos in small increments.
- Implemented DOM manipulation and timer-based state updates to achieve low-latency performance.

FTC Robotics | *Android Studio, OpenCV, TensorFlow, Autodesk Fusion*

Aug 2022 – Feb 2026

- Modeled the robot's core mechanisms in Autodesk Fusion, integrating principles of mechanical design and build constraints to optimize robot design.
- Some relevant designs were an induced-fit claw with three DoF and a spindexer coordinating with a rotating arm and a flywheel.

Technical Skills

Languages: Python, Java, C++, HTML/CSS, JavaScript, SQL

Developer Tools: VS Code, Eclipse, Amazon Web Services, Android Studio

Technologies/Frameworks: Linux, GitHub

Extracurricular

Running

Mid-Distance

Fall 2026 – Present

Massachusetts Institute of Technology

- 14:57 5K; 4:11 mile.